Expt 6 Finite Word length Effects

6.1 Uniform Quantization and Quantization Error Analysis of a Discrete-Time Sinusoidal Signal Using MATLAB

Objectives:

1. To generate a discrete-time sinusoidal signal in MATLAB.
2. To perform uniform quantization with different quantization levels.
3. To compare the original and quantized signals.
4. To analyze the quantization error and its effect on signal quality.

Procedure:

1. Open MATLAB and create a new script file.
2. Define the sampling frequency, signal frequency, and time vector.
3. Generate a discrete-time sinusoidal signal using the sine function.
4. Select different quantization levels such as 8, 16, and 32 levels.
5. Apply uniform quantization to the generated signal for each quantization level.
6. Compute the quantization error by subtracting the quantized signal from the original signal.
7. Calculate the Mean Square Error (MSE) to evaluate the quantization performance.
8. Plot the original signal and the quantized signals for comparison.
9. Plot the quantization error signals corresponding to each quantization level.
10. Analyze the effect of increasing quantization levels on signal quality and error reduction.

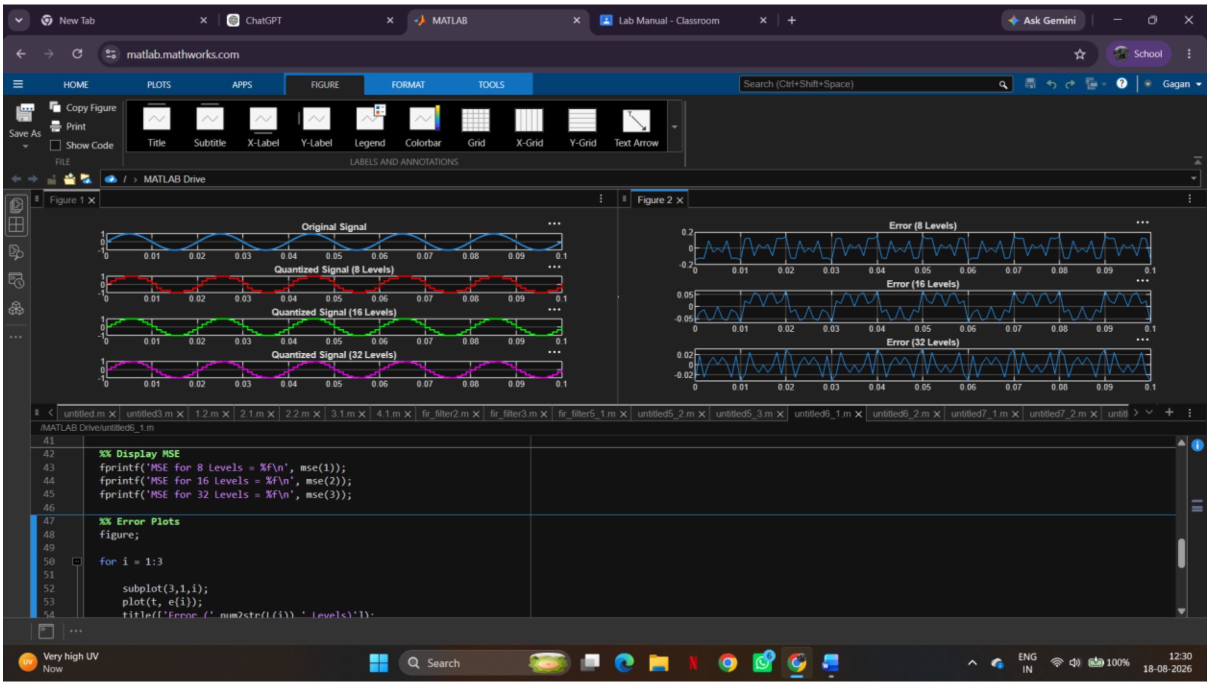
Program (Matlab code):

```
clc;

clear;

close all;


%% Signal

fs = 1000;

f = 50;

t = 0:1/fs:0.1;
```

```
x = sin(2*pi*f*t);

%% Quantization Levels
L = [8 16 32];
colors = {'r','g','m'};

figure;

for i = 1:length(L)

   %% Quantization
   xq{i} = round((x + 1) .* (L(i) - 1) / 2) ...
         .* (2 / (L(i) - 1)) - 1;

   %% Error & MSE
   e{i} = x - xq{i};
   mse(i) = mean(e{i}.^2);

   %% Plot Quantized Signals
   subplot(4,1,i+1);
   stairs(t, xq{i}, colors{i}, 'LineWidth', 1.2);
   title(['Quantized Signal (' num2str(L(i)) ' Levels)']);
   grid on;

end

%% Original Signal
subplot(4,1,1);
plot(t, x, 'LineWidth', 1.5);
title('Original Signal');
grid on;
```

```
%% Display MSE

fprintf('MSE for 8 Levels = %f\n', mse(1));

fprintf('MSE for 16 Levels = %f\n', mse(2));

fprintf('MSE for 32 Levels = %f\n', mse(3));


%% Error Plots

figure;


for i = 1:3


    subplot(3,1,i);

    plot(t, e{i});

    title(['Error (' num2str(L(i)) ' Levels)']);

    grid on;


end
```

Output:



Result:

1. A discrete-time sinusoidal signal was successfully generated in MATLAB.

2. Uniform quantization was performed for different quantization levels, and the corresponding quantized signals were obtained.

3. Comparison of the original and quantized signals showed that the quantized signal approximates the original signal more closely as the number of quantization levels increases.

4. The quantization error was analyzed and found to decrease with increasing quantization levels, resulting in improved signal quality and reduced quantization noise.